



Amrita School of AI
COIMBATORE CAMPUS

UNIT 1 · SESSIONS 1 AND 2

Complex numbers

The number system quantum mechanics is written in.

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SESSION 1

What a complex number is

Why we need one, what it is, and where it lives.

Every quantum state is a list of complex numbers.

By Session 11 you will write a qubit as two of them. Today: what one is.

Why real numbers are not enough

Quantum mechanics adds **amplitudes**, then squares the size of the sum to get a probability.

Complex amplitudes point in different directions, so two can cancel.

$$|z_1 + z_2|^2$$

Two routes to a detector can add up to **zero**.
Session 3 shows how.

$$x^2 = -1$$

No real number solves this. A real number squared is never negative.

$$i^2 = -1$$

Invent one new number, i , with this single property. Everything else is ordinary algebra.

A complex number

$$z = a + bi$$

Here a and b are ordinary real numbers.

The **real part**: $a = \operatorname{Re}(z)$

The **imaginary part**: $b = \operatorname{Im}(z)$

Example: $z = 3 - 2i$ has $\operatorname{Re}(z) = 3$ and $\operatorname{Im}(z) = -2$.

The imaginary part is a real number

- **Wrong**

$$\operatorname{Im}(3 - 2i) = -2i$$

The i marks the direction. It is not part of the number b .

- **Right**

$$\operatorname{Im}(3 - 2i) = -2$$

Both $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$ are real numbers.

The real numbers are already inside

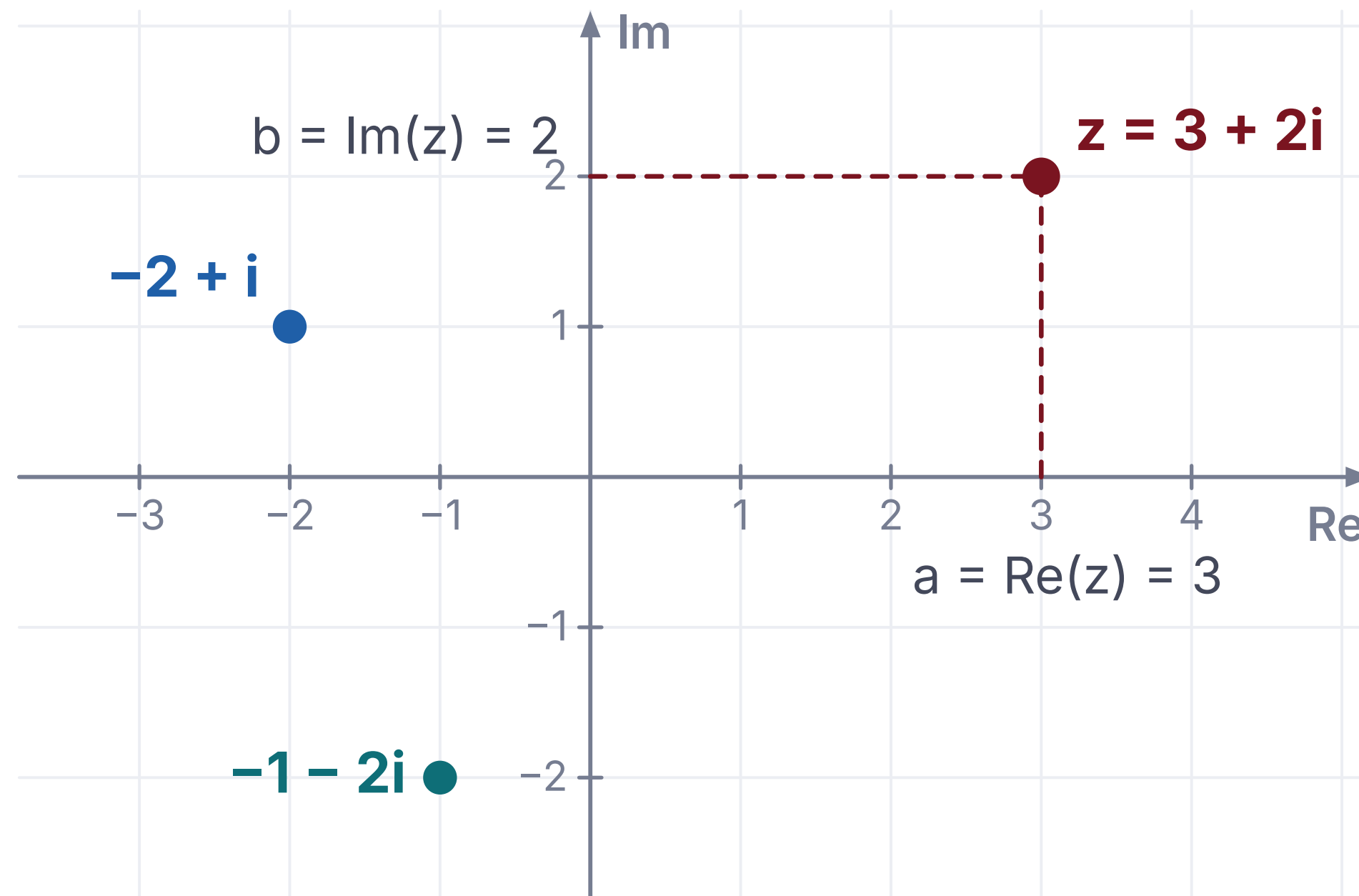
$$5 = 5 + 0i$$

A **real** number has $b = 0$.

$$4i = 0 + 4i$$

A **purely imaginary** number has $a = 0$.

$a + bi$ is the point (a, b)



One complex number = two real numbers.

So it needs a plane to live in, where a real number only needs a line.

Solve $x^2 + 4 = 0$

Move the 4 across and take square roots.

Use $\sqrt{-1} = i$ to handle the minus sign.

Check: $(2i)^2 = 4i^2 = -4$.

$$x^2 = -4$$

$$x = \pm\sqrt{-4}$$

$$= \pm\sqrt{4}\sqrt{-1}$$

$$= \pm 2i$$

Solve $x^2 - 2x + 5 = 0$

The quadratic formula still works:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

with $a = 1$, $b = -2$, $c = 5$.

$$\begin{aligned} x &= \frac{2 \pm \sqrt{4 - 20}}{2} \\ &= \frac{2 \pm \sqrt{-16}}{2} \\ &= \frac{2 \pm 4i}{2} \\ &= 1 \pm 2i \end{aligned}$$

Every quadratic now has two roots.

Complex roots come as a pair, $a + bi$ and $a - bi$. That pair gets a name in Session 2.

$\sqrt{a} \sqrt{b} = \sqrt{ab}$ **fails for negatives**

- **Wrong**

$$\sqrt{-4} \sqrt{-9} = \sqrt{36} = 6$$

- **Right**

$$\sqrt{-4} \sqrt{-9} = (2i)(3i) = -6$$

Turn each $\sqrt{-c}$ into $i\sqrt{c}$ **first**, then multiply.

Equal means both parts match

$$a + bi = c + di$$

This means $a = c$ **and** $b = d$. One complex equation gives two real equations.

Solve $(x + 1) + (y - 2)i = 4 + 3i$.

$$x + 1 = 4 \quad \Rightarrow \quad x = 3$$

$$y - 2 = 3 \quad \Rightarrow \quad y = 5$$

Plot these four numbers

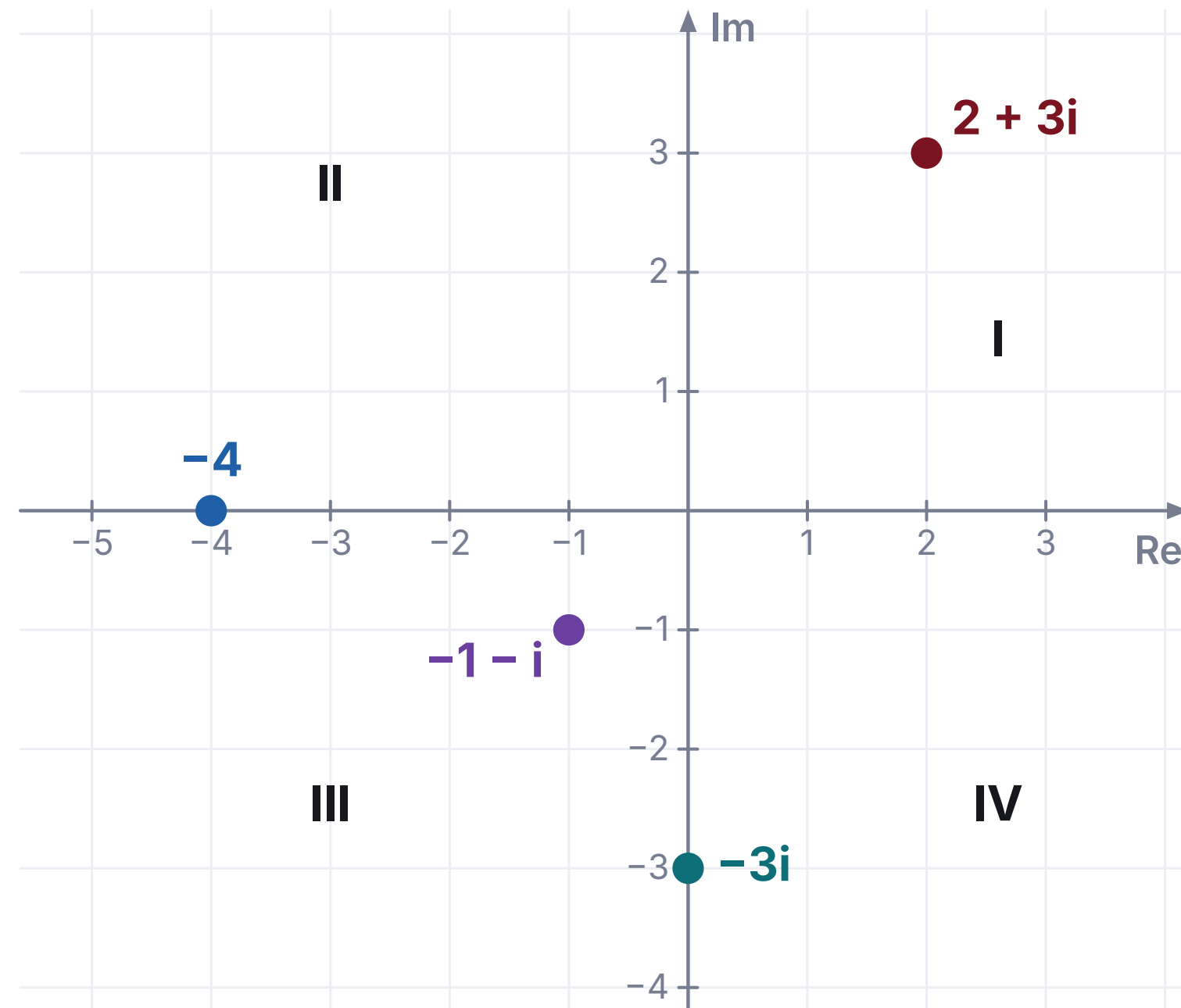
$$2 + 3i, \quad -4, \quad -3i, \quad -1 - i$$

Which quadrant is each one in? Which ones sit on an axis?

Then solve $x^2 + 9 = 0$.

Three minutes, on paper.

Where they sit



- $2 + 3i$: quadrant I
- $-1 - i$: quadrant III
- -4 : real axis
- $-3i$: imaginary axis
- $x^2 + 9 = 0$ gives $x = \pm 3i$

Python writes i as `1j`

```
z = 3 - 2j                # the number 3 - 2i
print(z.real, z.imag)     # 3.0 -2.0    (the imaginary part is real)
print(1j ** 2)            # (-1+0j)

import numpy as np
print(np.roots([1, -2, 5])) # roots of x^2 - 2x + 5: [1.+2.j 1.-2.j]
```

Engineers write j for i , and Python follows them. Type `1j`, never `j` alone.

Before Session 2

- Plot $1 + i$, $-2 + 3i$, $-5i$.
- Solve $x^2 + 6x + 13 = 0$.
- Find x and y if $2x + (x - y)i = 6 + i$.
- Open Google Colab (or Python on your laptop) and run `print((1j)**2)`.
- Read Laforest, Chapter 1, the section on what a complex number is.

SESSION 2

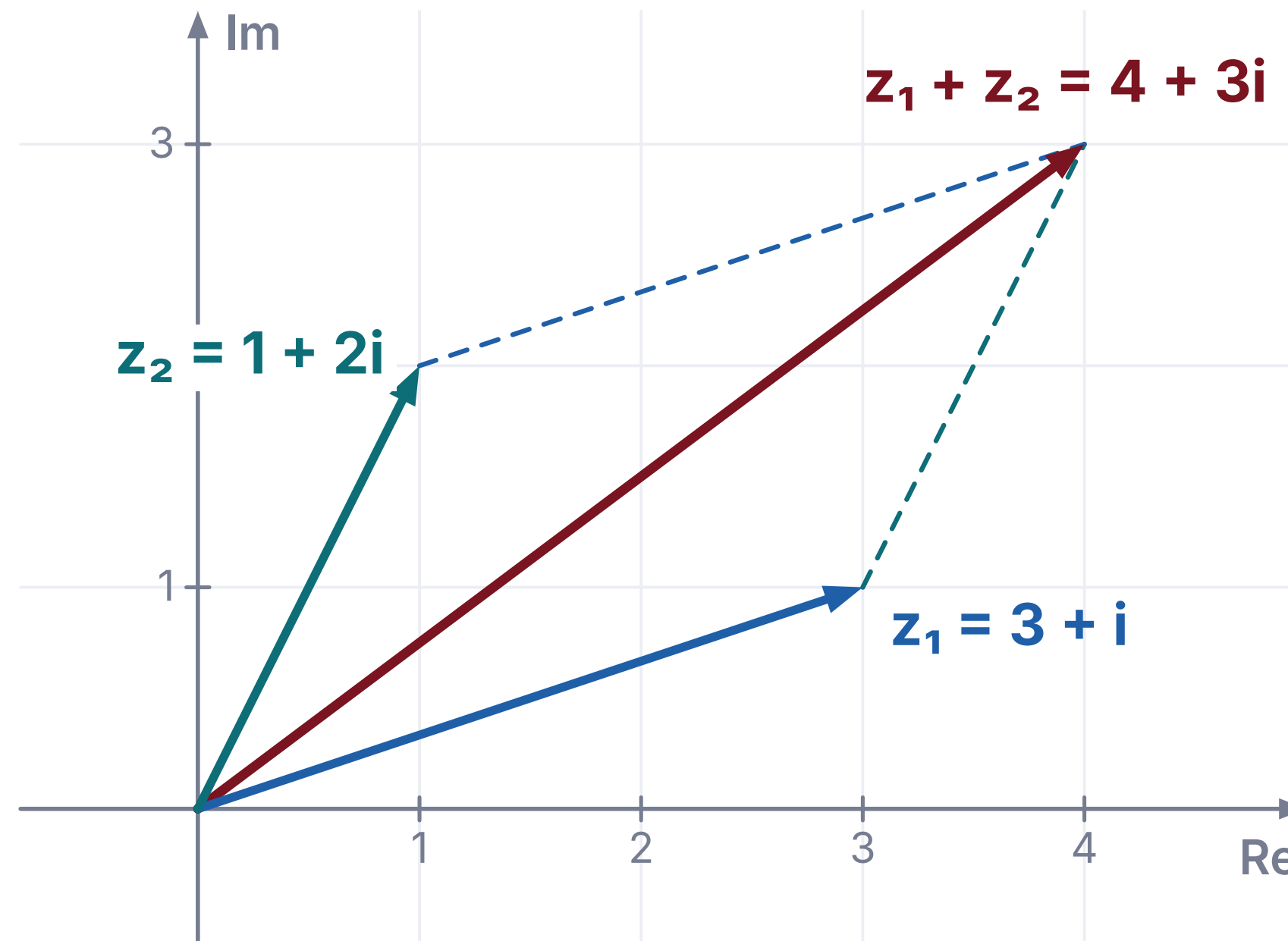
Doing arithmetic with complex numbers

Add, multiply, conjugate, measure the size, divide.

Treat i as a symbol. Replace i^2 by -1 .

That one rule is all the arithmetic in this session.

Adding is adding arrows



Real parts add. Imaginary parts add.

Part by part

$$\begin{aligned}(a + bi) + (c + di) \\ = (a + c) + (b + d)i\end{aligned}$$

$$\begin{aligned}(5 + 2i) - (1 - 3i) \\ = (5 - 1) + (2 + 3)i \\ = 4 + 5i\end{aligned}$$

Watch the sign: subtracting $-3i$ adds $3i$.

Multiply $(2 + 3i)(1 - 2i)$

Expand every term against every term, as you would for $(2 + 3x)(1 - 2x)$.

Then replace i^2 by -1 and collect.

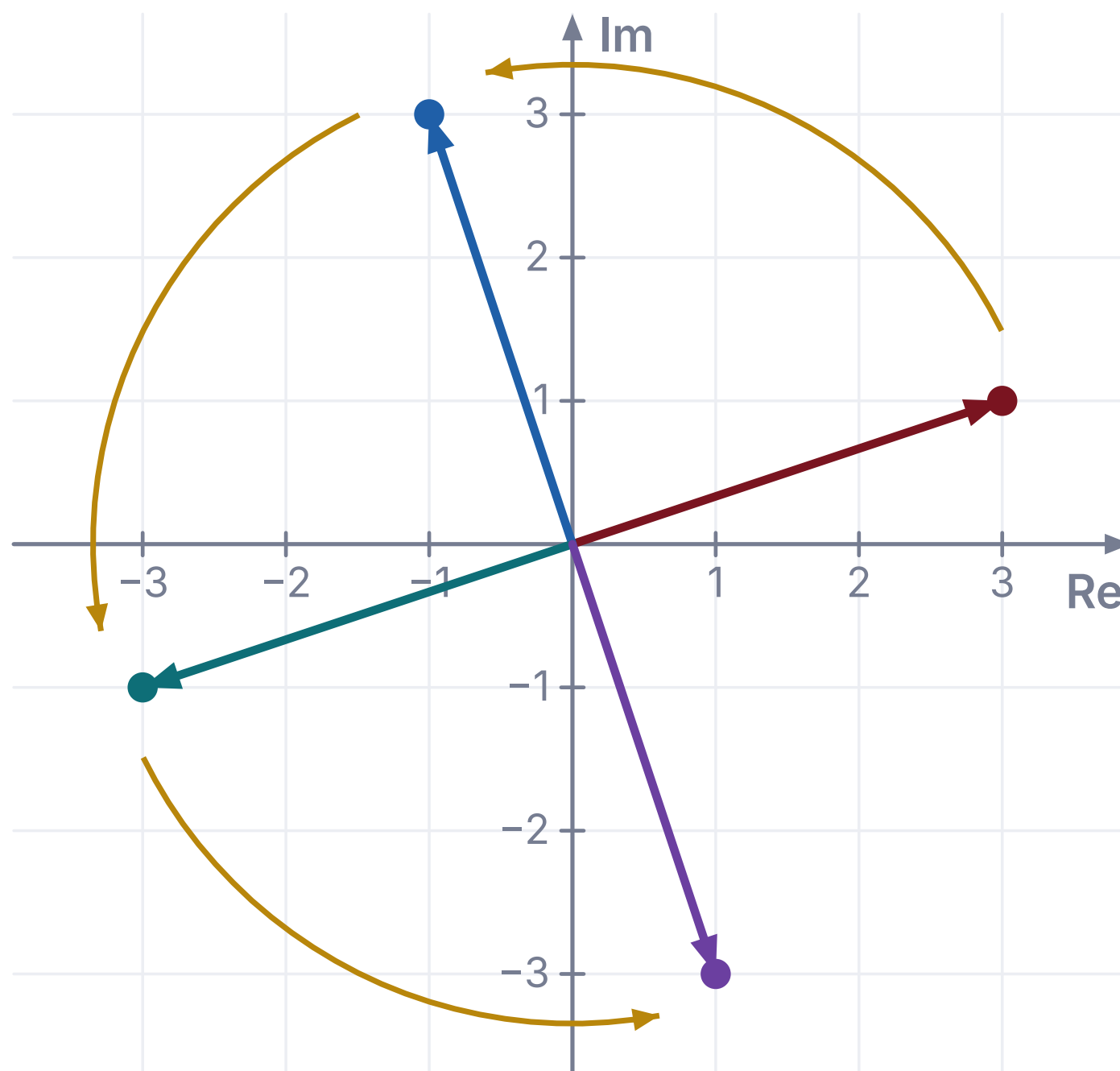
$$\begin{aligned}(2 + 3i)(1 - 2i) \\&= 2 - 4i + 3i - 6i^2 \\&= 2 - i + 6 \\&= 8 - i\end{aligned}$$

Multiplication, in general

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

Expanding each time is safer than memorising this.

Multiplying by i turns a point a quarter turn



$$z = 3 + i$$

$$iz = -1 + 3i$$

$$i^2z = -3 - i$$

$$i^3z = 1 - 3i$$

Each gold arc: $\times i$
a quarter turn, anticlockwise.
Length stays $\sqrt{10}$.

Powers of i repeat every four

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = i^2 \cdot i = -i$$

$$i^4 = (i^2)^2 = 1$$

Then $i^5 = i$ and the cycle starts again.

Find i^{27} . Since $27 = 4 \times 6 + 3$,

$$i^{27} = (i^4)^6 i^3 = -i$$

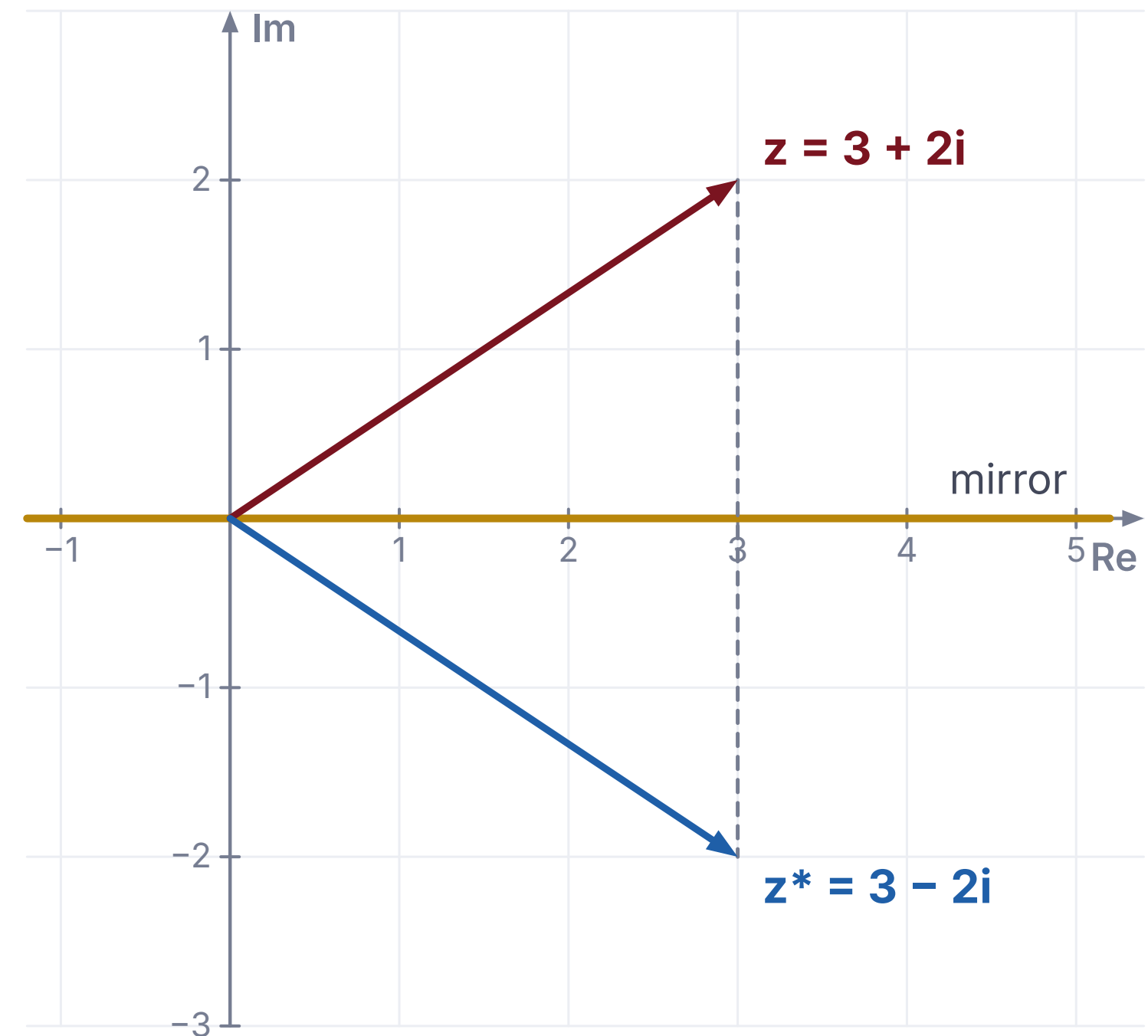
Divide the power by 4. Only the remainder matters.

Flip the sign of the imaginary part

$$z^* = a - bi$$

Read it "z star". Maths books write $\bar{z}$ for the same thing.

The roots $1 \pm 2i$ from Session 1 are conjugates.



A number times its conjugate

Take $z = 3 + 4i$, so $z^* = 3 - 4i$.

$$\begin{aligned} z^* z &= (3 - 4i)(3 + 4i) \\ &= 9 + 12i - 12i - 16i^2 \\ &= 9 + 16 = 25 \end{aligned}$$

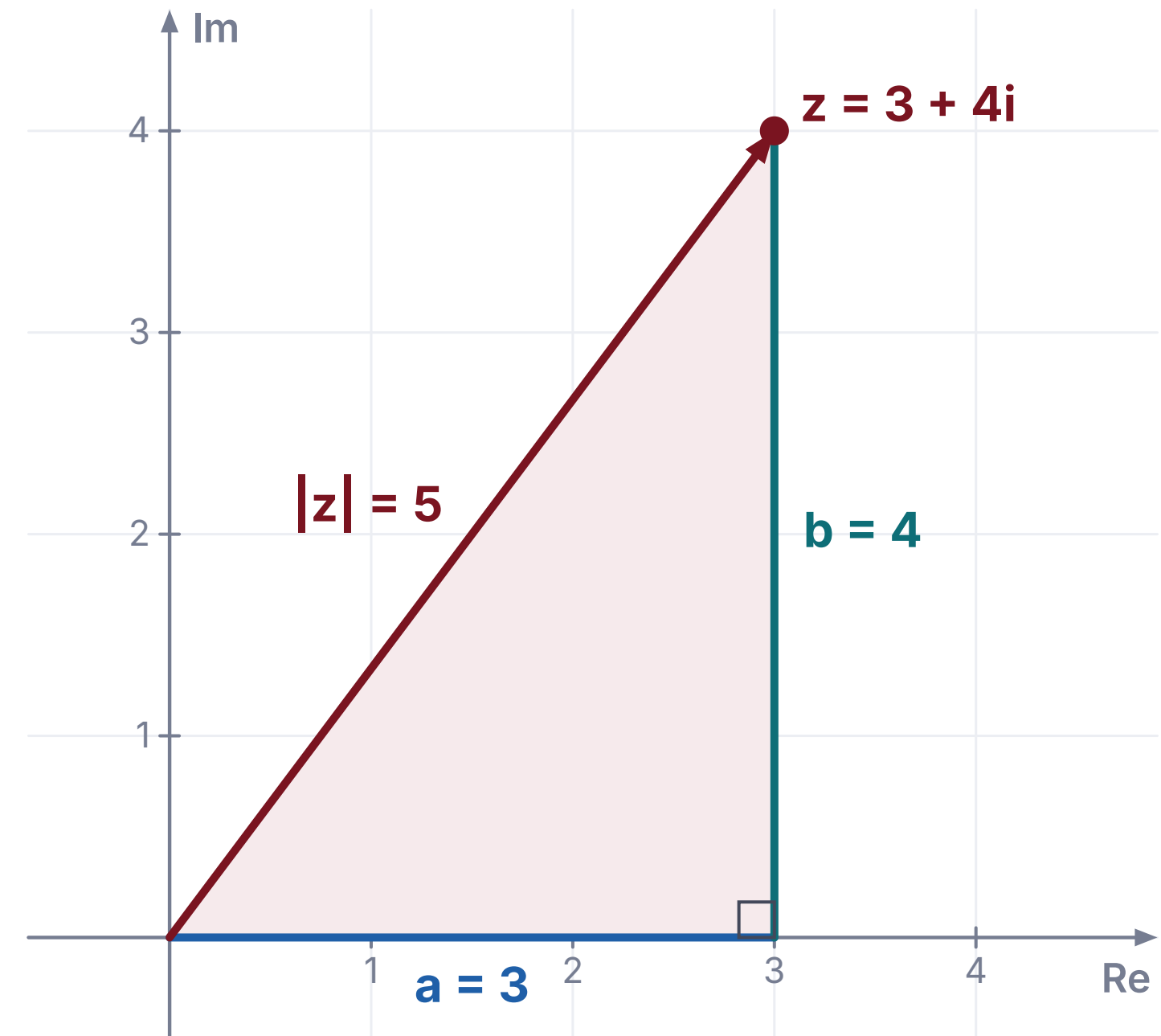
$$z^* z = a^2 + b^2$$

Always **real**, never negative, and zero only when $z = 0$.

The size of a complex number

$$|z| = \sqrt{z^* z} = \sqrt{a^2 + b^2}$$

The distance from 0 to the point (a, b) . Pythagoras, nothing more.



$|z|^2$ is not z^2

- **Squaring z**

$$\begin{aligned} z^2 &= (3 + 4i)^2 \\ &= 9 + 24i - 16 \\ &= -7 + 24i \end{aligned}$$

Complex, and can be anything.

- **Squaring the modulus**

$$|z|^2 = z^* z = 9 + 16 = 25$$

Real and never negative. Quantum probabilities are built from this one.

Clear the i out of the bottom

Multiply top and bottom by the **conjugate of the denominator**.

The bottom becomes $c^2 + d^2$, a real number, and the fraction splits.

$$\frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{c^2 + d^2}$$

Divide $5 + 5i$ by $1 + 2i$

The conjugate of the bottom is $1 - 2i$.

Check by multiplying back:

$$(3 - i)(1 + 2i) = 5 + 5i \checkmark$$

$$\begin{aligned}\frac{5 + 5i}{1 + 2i} &= \frac{(5 + 5i)(1 - 2i)}{(1 + 2i)(1 - 2i)} \\ &= \frac{5 - 10i + 5i + 10}{1 + 4} \\ &= \frac{15 - 5i}{5} = 3 - i\end{aligned}$$

What is $1/i$?

The conjugate of i is $-i$.

$$\frac{1}{i} = \frac{-i}{i(-i)} = \frac{-i}{-i^2} = \frac{-i}{1}$$

$$\frac{1}{i} = -i$$

Check: $i \cdot (-i) = -i^2 = 1.$

Conjugate and modulus behave well

$$(z_1 z_2)^* = z_1^* z_2^*$$

$$|z_1 z_2| = |z_1| |z_2|$$

$$z + z^* = 2 \operatorname{Re}(z)$$

$$z - z^* = 2i \operatorname{Im}(z)$$

$$z = 1 + i \text{ and } w = 2 - 3i$$

Find:

- $z + w$
- zw
- w^*
- $|w|$
- z/w

Five minutes. Check z/w by multiplying your answer by w .

$$z = 1 + i \text{ and } w = 2 - 3i$$

- $z + w = 3 - 2i$
- $zw = 2 - 3i + 2i - 3i^2 = 5 - i$
- $w^* = 2 + 3i$
- $|w| = \sqrt{4 + 9} = \sqrt{13}$

$$\begin{aligned}\frac{z}{w} &= \frac{(1 + i)(2 + 3i)}{(2 - 3i)(2 + 3i)} \\ &= \frac{2 + 3i + 2i - 3}{13} \\ &= \frac{-1 + 5i}{13}\end{aligned}$$

Hand first, then let Python confirm

```
import numpy as np

z, w = 1 + 1j, 2 - 3j
print(z + w, z * w)           # (3-2j) (5-1j)
print(np.conj(w), abs(w))     # (2+3j) 3.605551275463989 which is sqrt(13)
print(z / w)                  # (-0.0769...+0.3846...j) which is (-1 + 5i)/13
```

Here `abs(w)` is the modulus and `np.conj(w)` is the conjugate w^* .

Before Session 3

With $z = 2 + i$ and $w = 1 - i$, find zw , z/w , $|z|^2$ and i^{50} .

Check each one in Python.

Revise $\sin$ and $\cos$ of 0° , 30° , 45° , 60° , 90° . You need them next time.

Next: Euler's formula, and why multiplying by i is a turn.